

Collaborative Task Offloading in Space Computing Power Network: A World Model-Based Multi-Agent Reinforcement Learning Approach

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Abstract—Low Earth Orbit (LEO) constellations are required to process increasing volumes of heterogeneous tasks from ground networks. Intermittent inter-satellite links, heterogeneous onboard resources, and time-varying traffic loads make collaborative task offloading difficult for static or reactive strategies. Although multi-agent reinforcement learning (MARL) provides an adaptive solution, existing model-free MARL methods often suffer from slow convergence, insufficient foresight, and limited robustness in dynamic satellite environments. To address these challenges, this paper proposes a World Model-Based Multi-Agent Proximal Policy Optimization (WM-MAPPO) framework for space computing power networks. The offloading problem is formulated as a partially observable multi-agent decision-making process, where LEO satellites make decentralized decisions under incomplete local observations. A predictive world model learns latent transition dynamics of network states and provides future context for proactive planning. Meanwhile, a Transformer-based policy architecture captures inter-agent dependencies and supports cooperative scheduling under centralized training and decentralized execution (CTDE). Simulation results show that WM-MAPPO achieves higher task completion ratios, lower aver-

age latency, improved energy efficiency, and stronger robustness than model-free MARL baselines, heuristic methods, Lyapunov-based scheduling, and MINLP-inspired optimization.

Index Terms—Space computing power network, task offloading, multi-agent reinforcement learning (MARL), world model, proximal policy optimization (PPO).

I. INTRODUCTION

THE deployment of satellite constellations is reshaping global connectivity. An order-of-magnitude reduction in launch costs and advancements in satellite manufacturing have enabled a shift from single, dedicated satellites to large-scale, collaborative constellations operating across multiple orbits (LEO, GEO, SSO) for tasks such as communication, remote sensing (RS), and navigation [1], [2]. Increased on-orbit computational power enables the deployment of collaborative computing models [3]. Channel fading affects satellite-to-user links in addition to large-scale path loss. Fading models such as the λ - κ - μ distribution and the double λ - κ - μ distribution [4] provide additional flexibility for representing multipath and shadowing effects. As a result, multi-satellite cooperation and multi-task integration are becoming increasingly important [5], where the on-orbit task offloading problem arises for dynamically allocating computational loads and leveraging the collective resources of the constellation [6], [7].

To improve the computational capabilities and resource efficiency of multi-satellite networks, research has progressed through several offloading paradigms. Traditional strategies encompass a variety of optimization methodologies, such as heuristic optimization [8] and mixed-integer programming techniques that jointly optimize resource distribution and offloading decisions [9]. While traditional strategies offer theoretical rigor, they frequently face scalability bottlenecks and are unresponsive to the high dynamics of contemporary satellite constellations [10], [11]. Consequently, data-driven, learning-based methods have attracted substantial attention due to their adaptability in complex and dynamic environments [12], [13], [14], [15], [16], [17], leading to the adoption of deep reinforcement learning (DRL) for the adaptive policies. Early DRL approaches focused on enabling a single agent to make dynamic decisions. For instance, double Q-learning has been employed to manage traffic offloading in highly dynamic

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space-ground integrated networks by learning from local and historical network information [12]. To improve the stability of DRL performance, the authors in [18] used Dueling Double Deep Q-Networks (D3QN) for optimizing system delay. To further handle the scale of satellites, the authors in [14] have proposed distributed deep learning frameworks, where multiple parallel neural networks collaboratively improve sampling efficiency.

More recently, research has increasingly emphasized collaborative scheduling policies [19], and multi-agent reinforcement learning (MARL) has emerged as a more powerful and suitable paradigm [20]. By modeling each satellite as an autonomous agent, this approach naturally handles the scale and decentralized nature of satellite constellations [21], [22], [23], and often employs a centralized training and decentralized execution (CTDE) paradigm where agents learn a coordinated policy offline but execute it independently using only local information [20], [24]. The research has explored a variety of MARL algorithms to learn these sophisticated, decentralized control policies. These include value-based methods like QMIX using attention mechanisms to handle high-dimensional observations of inter-satellite link (ISL) states [22], actor-critic approaches such as Counterfactual Multi-Agent policy gradients (COMA) [20], and cooperative Multi-Agent Proximal Policy Optimization (MAPPO) [25]. These advanced methods make MARL the state-of-the-art approach for the satellite offloading problem [26], [27], [28]. In particular, the standard MAPPO algorithm has been reported to perform effectively in cooperative multi-agent tasks [29] and has been applied to multi-cell spectrum and power allocation [30], where each agent acts as an independent decision variable under the CTDE paradigm. The approach maps observations to actions without an explicit predictive component. Model-free methods remain reactive and do not explicitly represent deterministic topological changes in LEO satellite networks. Despite these advances, a fundamental limitation persists: the vast majority of current multi-agent learning frameworks are *reactive*. They excel at learning policies that map the current observation to an immediate action but lack a mechanism for foresight [31]. This reactive nature prevents them from anticipating predictable events, such as an upcoming high-bandwidth inter-satellite link window or a neighboring satellite's imminent availability, and planning ahead to exploit them.

To bridge this gap, a *proactive* planning approach based on a world model is introduced. Reactive agents cannot optimize decisions in a high-latency, partially observable environment when future link availability and congestion trends are not represented in the current observation. The proposed World Model-Based Multi-Agent Proximal Policy Optimization (WM-MAPPO) framework trains a predictive model of network dynamics. The model provides a latent forecast for lookahead planning and proactive offloading decisions. Simulation results indicate improved task completion ratio and resource utilization relative to the evaluated model-free MARL baselines. The main contributions are summarized as follows:

- A multi-layer (LEO, GEO, SSO) space-ground system model is specified. The dynamic task offloading problem

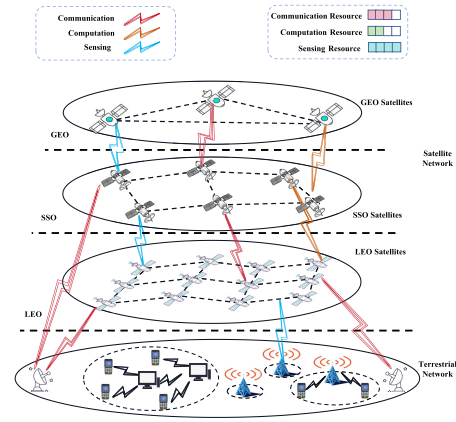


Fig. 1. The proposed multi-layered space-ground network architecture, illustrating the synergy between GEO and SSO remote-sensing satellites, a large constellation of LEO communication and computation satellites, and a diverse ground segment comprising ground stations.

is formulated as a Partially Observable Markov Decision Process (POMDP), representing decentralized decision-making under incomplete information.

- A WM-MAPPO framework is proposed. A predictive spatio-temporal world model is integrated to forecast future network states for lookahead planning. A Transformer-based policy architecture is applied to model inter-agent dependencies for cooperative scheduling.
- Comprehensive simulations using real-world data are conducted. The results indicate improved performance in completion ratio and latency relative to the evaluated model-free MARL baselines.

The remainder of this paper is organized as follows. Section II details the system model and problem formulation. Section III presents the proposed WM-MAPPO algorithm. Finally, Section IV validates the framework and Section V concludes the paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

This section first details the heterogeneous satellite network architecture, including the space and ground segments. The system models are then established, followed by the formulation of the task offloading problem.

A. System Architecture

As depicted in Fig. 1, a hierarchical space-ground integrated network architecture is considered, composed of four primary segments: a GEO Segment, an SSO Segment, an LEO Segment, and a Ground Segment. This multi-layered structure is designed to leverage the distinct advantages of different orbital altitudes and infrastructures.

1) *Ground Segment*: The Ground Segment serves as the vital interface between terrestrial service demands and the space-based infrastructure. This segment is composed of a set of geographically distributed Ground Stations, denoted by $\mathcal{G} = \{g_1, g_2, \dots, g_{|\mathcal{G}|}\}$. These stations function as the primary gateways for all terrestrial tasks. They are responsible for collecting a wide variety of task requests originating

from diverse end-users, integrating these demands, and then uplinking them to the satellite network via high-capacity links for subsequent processing.

2) *LEO Segment*: The LEO Segment is a massive constellation of satellites operating in Low Earth Orbit, forming a dynamic and responsive backbone for global data relay and edge computation. The satellites in this segment, denoted by $\mathcal{S}^{\text{leo}}$, are modeled after the Starlink constellation. Each satellite is multi-functional, equipped with both advanced communication payloads for a low-latency mesh network via ISLs and high-performance onboard processors that serve as orbital edge computing nodes. This integrated design allows the LEO segment to efficiently handle both communication-intensive and computation-intensive tasks.

3) *GEO Segment*: The GEO Segment consists of a set of satellites $\mathcal{S}^{\text{geo}}$ deployed in Geostationary Earth Orbit. Due to their high altitude and fixed relative positions, these satellites provide continuous, wide-area coverage, making them ideal for persistent monitoring and large-scale data acquisition. In addition to their primary remote-sensing function, they are equipped with communication capabilities, enabling them to act as high-altitude data relays, particularly for broadcasting data or offloading large volumes of collected data.

4) *Sso Segment*: The SSO Segment comprises a large constellation of remote-sensing satellites $\mathcal{S}^{\text{ssso}}$ in Sun-Synchronous Orbits. These satellites offer the advantage of consistent lighting conditions and regular revisit times over specific geographic locations. Complementing the persistent stare of the GEO segment, the SSO satellites are crucial for systematic, high-resolution Earth observation tasks.

The entire system can be modeled as a time-varying graph $G(t) = (\mathcal{V}, \mathcal{E}(t))$, where $\mathcal{V} = \mathcal{S}^{\text{geo}} \cup \mathcal{S}^{\text{ssso}} \cup \mathcal{S}^{\text{leo}} \cup \mathcal{G}$ is the set of all network resource nodes. End-users are the source of service demands that are aggregated at the ground stations but are not modeled as explicit vertices in the resource graph. $\mathcal{E}(t)$ is the set of active communication links (ISLs and space-ground links) at time t .

B. Communication Link Model

Communication links are essential for task offloading and data transmission. The viability and quality of a link between any two nodes $u, v \in \mathcal{V}$ depend on both the Line-of-Sight (LoS) condition and the received signal quality.

1) *Line-of-Sight and Non-Line-of-Sight Conditions*: A binary link state variable, $\xi_{u,v}(t) \in \{0, 1\}$, is introduced to represent the existence of a viable connection path. $\xi_{u,v}(t) = 1$ indicates an active LoS link, while $\xi_{u,v}(t) = 0$ signifies an NLoS or blocked link.

- For ISLs, a link is considered blocked if the line between the two satellites is obstructed by the Earth or another celestial body. This is a deterministic geometric condition.
- For satellite-to-ground links (SGLs), the connection is viable only if the satellite is above the ground node's horizon with a sufficient elevation angle to overcome local obstructions and minimize atmospheric attenuation. Let $\theta_{u,v}(t)$ be the elevation angle of satellite u as seen

from ground node v . A link exists only if $\theta_{u,v}(t) \geq \theta_{\min}$, where $\theta_{\min}$ is the minimum required elevation angle (e.g., 10°). The elevation angle can be calculated from the satellite's altitude h , the geocentric angle $\gamma_{u,v}(t)$ between ground node v and the subsatellite point of satellite u , and the Earth's radius R_E :

$$\theta_{u,v}(t) = \arctan \left(\frac{\cos(\gamma_{u,v}(t)) - \frac{R_E}{R_E + h}}{\sin(\gamma_{u,v}(t))} \right) \quad (1)$$

Thus, the link state is defined as:

$$\xi_{u,v}(t) = \begin{cases} 1, & \text{if LoS condition is met} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

2) *Received Signal Quality*: Given a viable LoS link, the data rate is determined by the received signal quality. The link model distinguishes between SGLs, which serve as the primary ingress for terrestrial tasks, and ISLs, which support in-orbit computation and data relay.

For the SGL, a link from a ground station u to a satellite v is deemed viable only if the received signal-to-noise ratio (SNR) surpasses a predefined quality threshold SNR_{th} . The SNR is derived from a detailed link budget incorporating antenna gains (G_u, G_v) and a comprehensive total path loss, $L^{\text{total}}(t)$. This loss term aggregates the standard Free-Space Path Loss (L^{FSPL}), a sophisticated atmospheric attenuation model (L^{atm}) based on ITU-R recommendations, and where applicable, sand and dust attenuation (L^{sand}) calculated using Mie scattering theory. Formally, the link admissibility is given by:

$$\mathcal{A}_{u,v}(t) = \begin{cases} 1, & \text{if } \xi_{u,v}(t) = 1 \text{ and } \text{SNR}_{u,v}(t) \geq \text{SNR}_{\text{th}} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where $\xi_{u,v}(t) = 1$ denotes LoS availability and $\text{SNR}_{u,v}(t)$ is the calculated SNR for the link in dB. A task is considered successfully uploaded only if $\mathcal{A}_{u,v}(t) = 1$.

For the ISL, the performance is governed by its link budget. In addition to the standard free-space path loss $L^{\text{FSPL}}(t)$ and fixed hardware implementation losses L^{hw} , a critical impairment for ISLs is the significant Doppler effect. The high relative velocities between LEO satellites induce a Doppler frequency shift $\Delta f(t)$, which can be calculated as:

$$\Delta f(t) = \frac{f_c \cdot v_{\text{rad}}(t)}{c} \quad (4)$$

where f_c is the carrier frequency and $v_{\text{rad}}(t)$ is the radial component of the relative velocity between the two satellites. This frequency offset can degrade the received SNR if it constitutes a substantial fraction of the channel bandwidth B . The degradation is modeled as an empirical SNR penalty term $L^{\text{Doppler}}(t)$, which is the normalized frequency shift:

$$L^{\text{Doppler}}(t) = 10 \log_{10} \left(1 + \left(\frac{2|\Delta f(t)|}{B} \right)^2 \right) \quad (5)$$

Synthesizing these elements, the final effective $\text{SNR}_{\text{eff}}(t)$ for ISL is explicitly calculated from the complete link budget, incorporating all relevant gains and losses:

$$\text{SNR}_{\text{eff}}(t) = P_u^{\text{tx}} + G_u + G_v - L^{\text{FSPL}}(t) - L^{\text{hw}} - L^{\text{Doppler}}(t) - P_N \quad (6)$$

where all terms are expressed in decibels. The achievable data rate is then determined by the Shannon-Hartley theorem, conditioned on LoS availability:

$$R_{u,v}(t) = \xi_{u,v}(t) \cdot B \log_2 \left(1 + 10^{\frac{\text{SNR}_{\text{eff}}(t)}{10}} \right) \quad (7)$$

Although Eq. (7) is written in a unified Shannon-rate form, its SNR and availability terms are evaluated according to the link type. For SGLs, $R_{g,s}(t)$ is obtained using the SGL admissibility indicator $\mathcal{A}_{g,s}(t)$ and the total path loss $L^{\text{total}}(t)$, including L^{FSPL} , L^{atm} , and, when applicable, L^{sand} ; for ISLs, $R_{u,v}(t)$ is computed from Eq. (6) with $L^{\text{FSPL}}(t)$, L^{hw} , and $L^{\text{Doppler}}(t)$. This distinguishes the SGL and ISL rate calculations without applying the same physical impairments to both link types.

C. Task Generation and Load Mapping Model

A task $k \in \mathcal{K}$ is characterized by the tuple $\mathcal{K}_k = \{\mathcal{T}_k, D_k, W_k, \tau_k^{\text{deadline}}, \mathcal{N}_k^{\text{src}}, t_k^{\text{arr}}\}$, defining its type, data size, workload, deadline, source, and arrival time. The task generation rate is driven by real-world geographical attributes. For each ground station $g \in \mathcal{G}$, its terrestrial service area is characterized by a feature vector, $\mathbf{F}_g = [F_g^{\text{pop}}, F_g^{\text{gdp}}, F_g^{\text{for}}, \dots]$, representing metrics such as population density, economic activity (GDP), and land use complexity. The demand for different task types is modeled as a function of different features. This is captured by a set of type-specific projection functions, $\{\text{proj}^{\mathcal{T}}(\cdot)\}$, where each function maps the geographical attributes to a specific task generation rate $\lambda_g^{\mathcal{T}}$ for type $\mathcal{T} \in \{\text{CM}, \text{CP}, \text{RS}\}$:

$$\lambda_g^{\mathcal{T}} = \text{proj}^{\mathcal{T}}(\mathbf{F}_g). \quad (8)$$

The aggregated task volume for each type from ground station g is then mapped to the LEO constellation, conditioned on the link admissibility defined in Section II-B. Let $\mathcal{S}_g^{\text{access}}(t)$ be the set of LEO satellites that have an admissible communication link with ground station g at time t :

$$\mathcal{S}_g^{\text{access}}(t) = \{s \in \mathcal{S}^{\text{leo}} \mid \mathcal{A}_{g,s}(t) = 1\} \quad (9)$$

In practical satellite networks, ground stations prioritize links with superior channel conditions to maximize uplink throughput. To capture this behavior, a channel-aware uplink scheduling mechanism is adopted. The probability $P_{g,s}(t)$ that ground station g offloads tasks to satellite $s \in \mathcal{S}_g^{\text{access}}(t)$ is proportional to the achievable data rate:

$$P_{g,s}(t) = \frac{R_{g,s}(t)}{\sum_{j \in \mathcal{S}_g^{\text{access}}(t)} R_{g,j}(t)}, \quad (10)$$

where $R_{g,s}(t)$ is the achievable data rate calculated via Eq. (7). This channel-aware mapping ensures that satellites with superior link budgets handle a larger proportion of the uplink traffic.

Consequently, the new data volume arriving at satellite s , denoted as $A_s(\Delta t)$, is given by:

$$A_s(\Delta t) = \sum_{g \in \mathcal{G}} \mathbb{I}(s \in \mathcal{S}_g^{\text{access}}(t)) \cdot P_{g,s}(t) \cdot \left(\sum_{\mathcal{T}} \lambda_g^{\mathcal{T}} \right) \Delta t, \quad (11)$$

where $\mathbb{I}(\cdot)$ is the indicator function and the summation over $\mathcal{T}$ represents the total task rate from station g . This realistic

mapping introduces dynamic traffic heterogeneity, driven by the time-varying relative positions of satellites and ground stations. The task queue $Q_s(t)$ on each satellite s evolves as:

$$Q_s(t + \Delta t) = [Q_s(t) - D_s^{\text{proc}}(\Delta t) - D_s^{\text{offload}}(\Delta t)]^+ + A_s(\Delta t) \quad (12)$$

where $D_s^{\text{proc}}(\Delta t)$ and $D_s^{\text{offload}}(\Delta t)$ are the data volumes processed locally and successfully offloaded during the time slot Δt , and $[\cdot]^+$ ensures the queue length is non-negative.

D. End-to-End Delay Model

Let $x_{u,v}^k(t) \in \{0, 1\}$ be a binary decision variable, where $x_{u,v}^k(t) = 1$ signifies that task k is offloaded from node u to node v at time t ; otherwise, $x_{u,v}^k(t) = 0$. The case where $u = v$ denotes local processing. For a given task k , the total completion time is an aggregation of delays incurred at various stages, including queuing, communication, and processing.

1) *Communication Delay*: The transmission delay is the time required to serialize the task's data onto the communication link. It is inversely proportional to the link's data rate $R_{u,v}(t)$ in (7):

$$T_{k,u,v}^{\text{tx}}(t) = \frac{D_k}{R_{u,v}(t)} \quad (13)$$

The propagation delay is the time it takes for the signal to travel the physical distance $d_{u,v}(t)$ between the nodes at the speed of light, c .

$$T_{u,v}^{\text{prop}}(t) = \frac{d_{u,v}(t)}{c} \quad (14)$$

The total one-hop communication delay is the sum of these two components, conditioned on the existence of a viable link ($\xi_{u,v}(t) = 1$).

$$T_{k,u,v}^{\text{comm}}(t) = T_{k,u,v}^{\text{tx}}(t) + T_{u,v}^{\text{prop}}(t). \quad (15)$$

2) *Computation Delay*: The computation delay is the time required for a satellite $s \in \mathcal{S}^{\text{cp}}$ to execute task k . This depends on the task's workload W_k and the computational resources allocated to it by the satellite, denoted as $f_{s,k}$. Let F_s be the total computational capacity of satellite s . The allocated resources must satisfy $\sum_k f_{s,k} \leq F_s$. The processing time is:

$$T_{k,s}^{\text{proc}} = \frac{W_k}{f_{s,k}}. \quad (16)$$

3) *Sensing Delay*: Each sensing task k specifies a minimum required precision, p_k^{req} . Correspondingly, each RS satellite $s \in \mathcal{S}^{\text{rs}}$ is equipped with a sensor that has a maximum precision capability, p_s^{cap} . For a satellite to be eligible, its sensor precision must meet or exceed the task's requirement. Furthermore, the accessibility of the target area is determined by the geometric visibility conditions. A satellite $s \in \mathcal{S}^{\text{rs}}$ can effectively sense the target area $\mathcal{A}_k$ at time t only if its elevation angle to all points $p \in \mathcal{A}_k$ remains above the minimum required threshold $\theta_{\min}$:

$$\text{is_accessible}(s, \mathcal{A}_k, t) \iff \theta_{s,p}(t) \geq \theta_{\min}, \quad \forall p \in \mathcal{A}_k \quad (17)$$

Therefore, the comprehensive condition for a satellite s to be capable of executing sensing task k at time t combines both precision and accessibility constraints:

$$\text{can_execute}(s, k, t) \iff (p_s^{\text{cap}} \geq p_k^{\text{req}}) \wedge \text{is_accessible}(s, \mathcal{A}_k, t) \quad (18)$$

The sensing delay is the time elapsed from the task's arrival until the visibility condition meets for a given duration:

$$T_{k,s}^{\text{sense}} = \min \left\{ t' \geq t_k^{\text{arr}} \mid \text{can_execute}(s, k, t) \forall t \in [t', t' + \tau_k^{\text{sense}}] \right\} - t_k^{\text{arr}} \quad (19)$$

4) *Queuing Delay*: The queuing delay for task k_{new} is the sum of the service times of all preceding tasks:

$$T_{k_{\text{new}},s}^{\text{queue}}(t_{\text{arr}}) = \sum_{k_i \in \mathbb{Q}_s(t_{\text{arr}})} T_{k_i,s} \quad (20)$$

where $T_{k_i,s}$ is the completion time for each task k_i .

5) *Total Task Completion Time*: The total completion time T_k for a task k originating at node $\mathcal{N}_k^{\text{src}}$ at time t_k^{arr} depends on the offloading decision. Let $\mathcal{P}_k = (n_0, n_1, \dots, n_m)$ be the selected path for task k , where $n_0 = \mathcal{N}_k^{\text{src}}$ is the source satellite that initially holds the task, and $n_m \in \mathcal{S}^{\text{cp}}$ is the final destination satellite that performs the computation.

- **Local Processing**: If the task is processed at its source node ($\mathcal{P}_k = (n_0)$), the total latency is the sum of the initial queuing delay and local processing delay:

$$T_k = T_{k,n_0}^{\text{queue}}(t_k^{\text{arr}}) + T_{k,n_0}^{\text{proc}} \quad (21)$$

- **Multi-hop Offloading**: If the task is routed along a path $\mathcal{P}_k$ of length $m > 0$, the total latency is the sum of communication delays at each hop, plus the final queuing and processing delay at the destination:

$$T_k = \sum_{i=0}^{m-1} T_{k,n_i,n_{i+1}}^{\text{comm}}(t_i) + T_{k,n_m}^{\text{queue}}(t_m) + T_{k,n_m}^{\text{proc}} \quad (22)$$

E. Energy Consumption Model

Since LEO satellites operate under strict onboard power constraints, evaluating energy consumption is essential for sustainable network operation. The total energy consumption for a satellite s comprises the processing of communication, computation, and sensing tasks.

1) *Communication Energy*: For a communication task k transmitted from node u to v , the energy consumption depends on the transmission power P_u^{tx} and the transmission time. Based on the data size D_k and the transmission rate $R_{u,v}(t)$ defined in (7), the communication energy is given by:

$$E_k^{\text{comm}} = P_u^{\text{tx}} \cdot \frac{D_k}{R_{u,v}(t)}. \quad (23)$$

2) *Computation Energy*: The task workload W_k (in CPU cycles) is assumed proportional to the data size D_k , expressed as $W_k = \rho \cdot D_k$, where ρ represents the computational complexity (cycles/bit). Following the standard CMOS dynamic power model [32], the computation energy for processing data size D_k is formulated as:

$$E_{k,s}^{\text{comp}} = \kappa \cdot (f_{s,k})^2 \cdot W_k = \kappa \rho D_k (f_{s,k})^2, \quad (24)$$

where κ is the effective switched capacitance coefficient depending on the chip architecture.

Utilizing the binary decision variable $x_{n_0,s}^k$ defined in Section II-D, where $n_0 = \mathcal{N}_k^{\text{src}}$ is the source node, the total energy consumption $E_k^{\text{totalcomp}}$ is expressed as:

$$E_k^{\text{totalcomp}} = \sum_{s \in \mathcal{S}^{\text{cp}}} x_{n_0,s}^k \left[\mathbb{I}(n_0 \neq s) \cdot P_{n_0}^{\text{tx}} \frac{D_k}{R_{n_0,s}(t)} + E_{k,s}^{\text{comp}} \right], \quad (25)$$

where the first term represents the upload energy cost if the target satellite s differs from the source n_0 , and the second term represents the computation energy cost.

3) *Sensing Energy*: Let P_s^{sense} denote the working power of the sensor on satellite s , and T_k^{dur} denote the required observation duration. The sensing energy is:

$$E_k^{\text{sense}} = P_s^{\text{sense}} \cdot T_k^{\text{dur}}. \quad (26)$$

The total energy consumption for satellite s at time slot t is the summation of energy consumed by all assigned tasks:

$$E_s^{\text{total}}(t) = \sum_{k \in \mathcal{K}_s^{\text{comm}}} E_k^{\text{comm}} + \sum_{k \in \mathcal{K}_s^{\text{comp}}} E_k^{\text{totalcomp}} + \sum_{k \in \mathcal{K}_s^{\text{sense}}} E_k^{\text{sense}}. \quad (27)$$

4) *Average Effective Energy Consumption*: To evaluate the energy efficiency, the metric of average effective energy consumption (AEEC) is introduced. Let $\mathcal{K}_{\text{succ}}$ and $\mathcal{K}_{\text{fail}}$ denote the sets of successfully completed and failed tasks within a given observation period, respectively. The AEEC is defined as the ratio of total system energy consumption $E_{\text{sys}}^{\text{total}}$ to the number of successfully completed tasks $|\mathcal{K}_{\text{succ}}|$:

$$E_{\text{eff}}^{\text{avg}} = \frac{\sum_{k \in \mathcal{K}_{\text{succ}} \cup \mathcal{K}_{\text{fail}}} (E_k^{\text{comm}} + E_k^{\text{totalcomp}} + E_k^{\text{sense}})}{|\mathcal{K}_{\text{succ}}|}. \quad (28)$$

A lower $E_{\text{eff}}^{\text{avg}}$ implies that the system consumes less energy to deliver a unit of valid service.

F. Problem Formulation

The opportunistic task offloading problem in the space-ground integrated network is formulated as a joint optimization of task routing and resource allocation. The objective is to minimize the total weighted completion time for all tasks by determining the optimal routing path $\mathcal{P}_k$ and computational resource allocation $f_{s,k}$ for each task $k \in \mathcal{K}$. This can be expressed as the following optimization problem:

$$\min_{\mathbf{X}, \mathbf{f}} \sum_{k \in \mathcal{K}} T_k(\mathbf{X}, \mathbf{f}) \quad (29a)$$

$$\text{s.t.} \quad \sum_{s \in \mathcal{S}^{\text{cp}}} x_{s,s}^k = 1, \quad \forall k \in \mathcal{K} \text{ with } \mathcal{T}_k = \text{CP} \quad (29b)$$

$$\sum_{s \in \mathcal{S}^{\text{rs}}} x_{s,s}^k = 1, \quad \forall k \in \mathcal{K} \text{ with } \mathcal{T}_k = \text{RS} \quad (29c)$$

$$\sum_{k \in \mathcal{K}} x_{s,s}^k f_{s,k} \leq F_s, \quad \forall s \in \mathcal{S}^{\text{cp}} \quad (29d)$$

$$\sum_{k \in \mathcal{K}} P_{s,k}^{\text{task}} \leq P_s^{\text{max}}, \quad \forall s \in \mathcal{S}^{\text{cp}} \quad (29e)$$

$$T_k(\mathbf{X}, \mathbf{f}) \leq \tau_k^{\text{deadline}}, \quad \forall k \in \mathcal{K} \quad (29f)$$

$$x_{u,v}^k \leq \xi_{u,v}(t), \quad \forall k \in \mathcal{K}, \quad \forall u, v \in \mathcal{V}, u \neq v \quad (29g)$$

$$x_{u,v}^k \in \{0, 1\}, \quad \forall k \in \mathcal{K}, \forall u, v \in \mathcal{V} \quad (29h)$$

$$f_{s,k} \geq 0, \forall k \in \mathcal{K}, \forall s \in \mathcal{S}^{\text{cp}} \quad (29i)$$

where $\mathbf{X} = \{x_{u,v}^k\}$ and $\mathbf{f} = \{f_{s,k}\}$ are the sets of all decision variables. The objective (29a) is to minimize the total task completion time. Constraints (29b) and (29c) ensure that each computation or sensing task is assigned to exactly one satellite with the appropriate capability. Constraint (29d) guarantees that the total allocated computational resources on a satellite do not exceed its capacity F_s . Constraint (29e) limits the scheduled task power on satellite s , where $P_{s,k}^{\text{task}}$ is the instantaneous task power and P_s^{max} is the onboard power budget. Constraint (29f) enforces the task-specific deadlines τ_k^{deadline} . Constraint (29g) ensures that routing only occurs over available LoS links (where $\xi_{u,v}(t)$ is 1 at the time of transmission). Finally, constraints (29h) and (29i) define the domains for the decision variables. Together with Section II-E and the energy-aware reward, this makes the formulation power-feasible and energy-aware.

III. WM-MAPPO: A WORLD MODEL-BASED MAPPO ALGORITHM FOR SATELLITE TASK OFFLOADING

To address the NP-hard problem in dynamic satellite networks, a model-based multi-agent reinforcement learning framework is formulated and a WM-MAPPO algorithm is proposed.

A. POMDP Formulation

The problem contains the task offloading part and resource allocation part. To solve the complex mixed-integer programming problem in an intelligent and adaptive manner, the offloading part is modeled as a POMDP, defined by the tuple $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}_{i \in \mathcal{N}}, \{\Omega_i\}_{i \in \mathcal{N}}, \mathcal{T}, \mathcal{R} \rangle$, where $\mathcal{N}$ is the set of N agents (LEO satellites, $s \in \mathcal{S}^{\text{leo}}$).

1) *State Space ($\mathcal{S}$)*: The global state $s_t \in \mathcal{S}$ at time t provides a comprehensive snapshot of the network. It is a composite tuple $s_t = (\mathbf{Q}_t, \mathbf{R}_t, \mathbf{F}_t)$:

- $\mathbf{Q}_t = \{Q_s(t) \mid s \in \mathcal{S}^{\text{leo}}\}$: The set of all task queues for each agent (satellite), where each $Q_s(t)$ contains tasks with attributes $\{\mathcal{T}_k, D_k, W_k, \tau_k^{\text{deadline}}\}$.
- $\mathbf{R}_t$: An $N \times N$ matrix representing network link quality, where element $R_{u,v}(t)$ is the data rate between satellites u and v .
- $\mathbf{F}_t$: An N -dimensional vector where $F_s(t)$ is the remaining computational capacity of satellite $s \in \mathcal{S}^{\text{cp}}$.

2) *Observation Space (Ω_i)*: Each agent i perceives a local observation $o_{i,t} \in \Omega_i$, including its own state and that of its one-hop neighbors $\mathcal{N}_i(t)$:

$$o_{i,t} = [\mathbf{q}_{i,t}, \mathbf{f}_{i,t}, \{(\mathbf{r}_{i,j,t}, \mathbf{f}_{j,t})\}_{j \in \mathcal{N}_i(t)}] \quad (30)$$

where $\mathbf{q}_{i,t}$ is agent i 's task queue state, $\mathbf{f}_{i,t}$ is its resource vector, and $\mathbf{r}_{i,j,t}$ is the data rate to neighbor j .

3) *Action Space ($\mathcal{A}_i$)*: The action $a_{i,t} \in \mathcal{A}_i$ for agent i is the offloading decision for a given task. The action space is discrete, containing the agent itself (local processing) and all its current neighbors: $\mathcal{A}_i(t) = \{i\} \cup \mathcal{N}_i(t)$.

4) *Reward Function ($\mathcal{R}$)*: A shared reward function is employed to drive agents toward the global objective. The goal is to maximize the task completion rate while minimizing latency and energy consumption:

$$r_t = w_{\text{succ}} N_t^{\text{succ}} - w_{\text{lat}} \sum_{k \in \mathcal{K}_t^{\text{comp}}} T_k - w_{\text{eng}} \sum_{s \in \mathcal{S}^{\text{leo}}} E_s^{\text{total}}(t) - w_{\text{fail}} N_t^{\text{fail}} \quad (31)$$

where N_t^{succ} and N_t^{fail} are the numbers of successfully completed and failed tasks, respectively. The term $\sum T_k$ is the sum of their completion times. The term $\sum E_s^{\text{total}}(t)$ represents the total energy consumption of the constellation at time t , as defined in Eq. (27). The coefficients w_{succ} , w_{lat} , w_{eng} , and w_{fail} balance these objectives.

To ensure stable learning across heterogeneous tasks, a running reward normalization scheme is applied. The raw reward r_t is scaled by its running standard deviation σ_r :

$$\hat{r}_t = \frac{r_t}{\sigma_r + \epsilon} \quad (32)$$

where $\epsilon = 10^{-8}$ is a small constant for numerical stability. This scaling mechanism ensures that the advantage function $\hat{A}_t^{\text{GAE}}$ remains within a consistent range, preventing gradient explosion or vanishing caused by extreme latency values or task failure penalties.

B. Model-Based MARL for Proactive Planning

While the POMDP provides a formal problem structure, solving it effectively with standard model-free MARL algorithms presents two major hurdles. First, these methods are purely reactive, learning a direct mapping from observations to actions without any inherent capability for long-term planning. In the satellite environment where future opportunities like high-bandwidth connectivity windows are predictable, this reactive nature is suboptimal. Second, model-free methods are often sample-inefficient, requiring a vast number of interactions with the real environment, which is costly and time-consuming in a complex satellite network.

To overcome these limitations, a predictive world model $\mathcal{M}_\phi$ is integrated. This model learns the dynamics of the environment, enabling proactive lookahead planning and improving sample efficiency by serving as a simulator for policy training.

1) *Learning System Dynamics From Environment*: The world model $\mathcal{M}_\phi$ is explicitly trained to learn the *temporal transition dynamics* of the satellite network, approximating the state transition function $\mathcal{P}(s_{t+1} | s_t, \mathbf{a}_t)$. This is fundamentally distinct from a standard latent feature encoder that merely compresses static observations into a fixed representation.

The world model predicts the next state embedding $\mathbf{z}_{t+1}$ given the current global state s_t and the joint action $\mathbf{a}_t$:

$$\mathbf{z}_{t+1} = \mathcal{M}_\phi(s_t, \mathbf{a}_t), \quad (33)$$

where $\mathbf{z}_{t+1} \in \mathbb{R}^{d_z}$ is the predicted latent representation of the next global network state. Since $\mathcal{M}_\phi$ is conditioned on both s_t and the joint action $\mathbf{a}_t$, it models action-dependent transitions rather than only static feature encoding.

In implementation, the global state is vectorized from task queues, link qualities, and satellite resource states as:

$$s_t = [\mathbf{q}_t, R_t, F_t] \in \mathbb{R}^{N_s \times d_s}, \quad (34)$$

where each row contains the per-satellite queue, link-quality, and resource features. The joint action $\mathbf{a}_t$ collects the offloading decisions of all satellites, and the latent dimension is set to $d_z = 256$ in the experiments.

The world model is trained on environment trajectories by minimizing the one-step latent MSE:

$$\mathcal{L}_{\text{WM}}(\phi) = \frac{1}{|\mathcal{D}|} \sum_{(s_t, \mathbf{a}_t, s_{t+1}) \in \mathcal{D}} \|\mathcal{M}_\phi(s_t, \mathbf{a}_t) - \mathbf{z}_{t+1}^{\text{true}}\|_2^2, \quad (35)$$

where $\mathcal{D}$ is the transition dataset and $\mathbf{z}_{t+1}^{\text{true}}$ is encoded from the observed next state s_{t+1} . This objective makes $\mathcal{M}_\phi$ a learned latent transition model for lookahead planning.

2) *Model-Based Proactive Planning*: The planning mechanism employed is *Latent Imagination*: the model performs a lookahead rollout in the compact latent space over a finite horizon H to generate a future context vector, which then conditions the policy network. This approach avoids the computational cost of model predictive control (MPC) with online trajectory optimization, which is prohibitive for onboard satellite processing.

- **Predict Future Context**: At each decision step t , the agent uses $\mathcal{M}_\phi$ to forecast the most likely future states by rolling out a trajectory for horizon H using the current policy π_θ :

$$\begin{aligned} \hat{\mathbf{z}}_{t+h+1} &= \mathcal{M}_\phi(\hat{s}_{t+h}, \mathbf{a}_{t+h}), \\ \mathbf{a}_{t+h} &\sim \pi_\theta(\cdot | \hat{s}_{t+h}), \quad h = 0, \dots, H-1, \quad \hat{s}_t = s_t \end{aligned} \quad (36)$$

Here, H is the finite prediction horizon for latent lookahead, and $\epsilon_\tau = \hat{\mathbf{z}}_{\tau+1} - \mathbf{z}_{\tau+1}^{\text{true}}$ denotes the latent prediction error, which may accumulate as the rollout horizon increases.

Under a locally Lipschitz value approximation, the induced deviation in the H -step return can be bounded as

$$|\Delta J_H| \lesssim \sum_{\tau=0}^{H-1} \gamma^\tau L_J \|\epsilon_\tau\|_2, \quad (37)$$

where L_J denotes the local sensitivity to latent prediction errors. This reflects the tradeoff between longer lookahead and error accumulation.

- **Informed Policy Decision**: The predicted latent rollout is summarized as a future context vector $\bar{\mathbf{z}}_{t,H}$, which is concatenated with the current local observation $o_{i,t}$ to form an augmented observation:

$$o'_{i,t} = [o_{i,t}, \bar{\mathbf{z}}_{t,H}]. \quad (38)$$

The policy network then selects an action conditioned on this forward-looking input:

$$a_{i,t} \sim \pi_\theta(\cdot | o'_{i,t}). \quad (39)$$

C. World Model-Based MAPPO Offloading Framework

In this part, the Proximal Policy Optimization (PPO) algorithm is introduced, followed by MAPPO and the proposed WM-MAPPO offloading framework.

1) *Proximal Policy Optimization*: PPO [33] has emerged as a leading algorithm in deep reinforcement learning due to its implementation simplicity and robust performance. It improves upon traditional policy gradient methods, which are often sensitive to step size and can suffer from catastrophic policy collapse, by constraining the magnitude of policy updates. PPO operates within an actor-critic framework. The *actor* is the policy network, denoted by $\pi_\theta(a|s)$, which is parameterized by θ and maps a state s to a distribution over actions a . The *critic* is the value network, $V_\phi(s)$, parameterized by ϕ , which estimates the expected return (the state-value) from state s . The critic's role is to reduce the variance of the policy gradient estimates.

The policy update is guided by the advantage function, $A^\pi(s, a)$, which quantifies how much better a given action a is compared to the average action in a state s , defined as:

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s), \quad (40)$$

where $Q^\pi(s, a)$ is the action-value function under policy π . In practice, the critic's output $V_\phi(s)$ is used as an estimate for $V^\pi(s)$ and the temporal-difference (TD) error is used as a one-step estimate for the advantage:

$$\hat{A}_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \quad (41)$$

where r_t is the reward at time t and $\gamma \in [0, 1)$ is the discount factor. For a more stable, lower-variance estimate, the generalized advantage estimation (GAE) is employed:

$$\hat{A}_t^{\text{GAE}} = \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}, \quad (42)$$

where $\lambda \in [0, 1]$ is a parameter for balancing bias and variance, and $\delta_{t+k} = r_{t+k} + \gamma V_\phi(s_{t+k+1}) - V_\phi(s_{t+k})$ is the TD error at time step $t+k$.

The key innovation of PPO is its clipped surrogate objective function, which constrains the policy update to a small trust region around the old policy. This prevents overly aggressive updates that could destabilize the learning process. The objective is defined as:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]. \quad (43)$$

The components of this objective are:

- **Probability Ratio**: $\rho_t(\theta)$ is the ratio of the probability of an action under the current policy π_θ to that under the old policy $\pi_{\theta_{\text{old}}}$ used to collect the data:

$$\rho_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}. \quad (44)$$

- **Clipping Function**: The $\text{clip}(\cdot)$ function constrains the probability ratio to the interval $[1 - \epsilon, 1 + \epsilon]$, where ϵ is a small hyperparameter (e.g., 0.2).

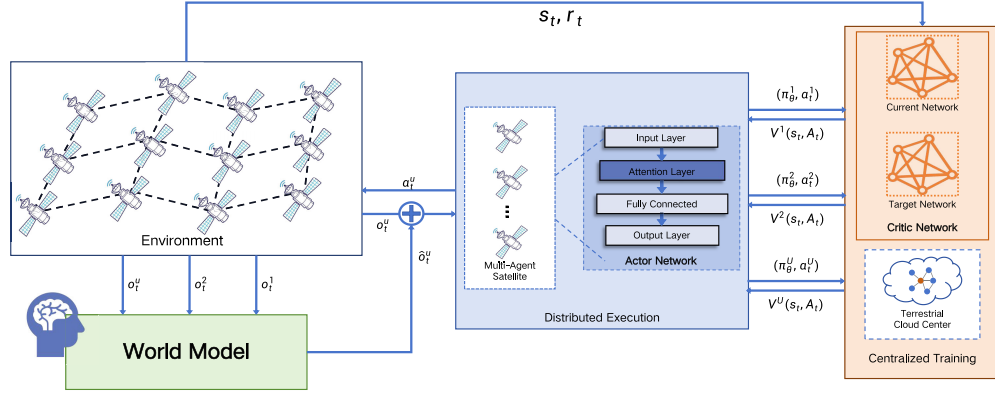


Fig. 2. The architecture of the proposed WM-MAPPO framework for satellite task offloading.

2) *Transformer-Based MAPPO Design*: In the satellite network, multiple satellites operate concurrently, making it a multi-agent system. Directly applying independent PPO to each agent is often ineffective due to the non-stationarity of the environment from any single agent's perspective. MAPPO [29] addresses this by extending PPO using the CTDE paradigm.

- **Centralized Critic**: The critic $V_\phi(s_t)$ is trained on the global state s_t . Its value loss is:

$$L^{VF}(\phi) = \mathbb{E}_t [(V_\phi(s_t) - G_t)^2], \quad (45)$$

where G_t is the empirical return. The GAE for each agent i is then computed using this centralized value estimate, providing a consistent signal for all agents.

- **Decentralized Actors**: Each actor π_{θ_i} learns its policy based on its local observation $o_{i,t}$. The objective for the entire multi-agent system is the average of the individual objectives:

$$L_{\text{MAPPO}}^{\text{CLIP}}(\theta) = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_t \left[\min(\rho_{i,t}(\theta_i) \hat{A}_{i,t}^{\text{GAE}}, \text{clip}(\rho_{i,t}(\theta_i), 1 - \epsilon, 1 + \epsilon) \hat{A}_{i,t}^{\text{GAE}}) \right]. \quad (46)$$

In this paper, parameter sharing is used for the actor networks ($\theta_i = \theta$ for all i), which enhances sample efficiency and promotes coordinated behavior. By leveraging a centralized critic, MAPPO effectively stabilizes the learning process in the multi-agent setting, making it a powerful and suitable algorithm for optimizing the complex task offloading policies.

3) *WM-MAPPO Framework for Task Offloading*: In satellite multi-agent systems, task offloading is a critical mechanism for executing computationally intensive tasks by leveraging distributed resources. The primary challenges lie in the highly dynamic network topology, intermittent connectivity, and agent heterogeneity. As shown in Fig. 2, the proposed WM-MAPPO framework comprises a world model that learns the environmental transition dynamics and a multi-agent policy layer that executes collaborative offloading.

As the initial step, each agent i (e.g., a satellite) identifies a set of candidate nodes $C_{i,t}$ for potential task offloading. This set includes neighboring satellites within its communication

range, selected based on link quality and availability. Subsequently, agent i formulates its local observation $o_{i,t}$ and the relevant states of its candidates in $C_{i,t}$.

The internal structure of the policy network is depicted in Fig. 3. A Transformer encoder is employed as the implicit world model component. Its purpose is to process the joint local observations $\mathbf{o}_t$ from all agents and produce a globally-aware, predictive latent state representation. Each raw local observation $o_{i,t}$ is mapped into a dense feature vector using an embedding network:

$$h_{i,t} = \text{Embed}(o_{i,t}; \theta_{\text{emb}}). \quad (47)$$

The resulting sequence of embeddings, $\mathbf{h}_t = (h_{1,t}, \dots, h_{N,t})$, is then fed into the encoder. Through its multi-head self-attention mechanism, the encoder infers complex spatio-temporal interdependencies among all agents.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V, \quad (48)$$

$$\text{MultiHead}(\mathbf{h}_t) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) W^O. \quad (49)$$

The output sequence $\mathbf{z}_t$ is the latent world state, which captures the distilled system dynamics. It is then distributed to all agents to guide their individual decisions.

4) *WM-MAPPO With CTDE Paradigm*: A Transformer decoder generates the joint action $\mathbf{a}_t = (a_{1,t}, \dots, a_{N,t})$ in an auto-regressive fashion, tractably factorizing the joint policy:

$$\pi_\theta(\mathbf{a}_t | \mathbf{o}_t) = \prod_{i=1}^N \pi_\theta(a_{i,t} | \mathbf{o}_t, \mathbf{a}_{< i,t}), \quad (50)$$

where $\mathbf{a}_{< i,t}$ represents the actions of preceding agents. To generate action $a_{i,t}$, the decoder attends to both the global context $\mathbf{z}_t$ (via the world model encoder) and the generated actions $\mathbf{a}_{< i,t}$ of other agents (via masked self-attention). The decoder's final output layer produces the probability distribution for action $a_{i,t}$.

To strictly adhere to the CTDE paradigm, each agent i maintains a local instance of the pre-trained world model parameters during execution. At each time step t , agent i processes its local observation sequence $o_{i,t}$ through this local encoder to generate a locally inferred latent belief

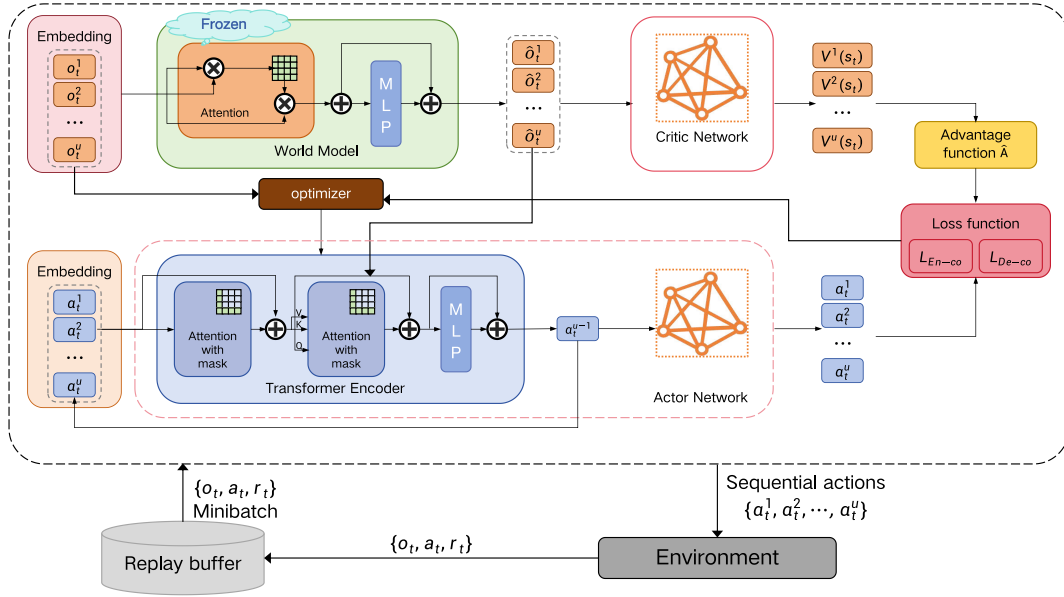


Fig. 3. The architecture of the Transformer-based policy network (actor) for multi-agent sequential decision-making under the CTDE paradigm, where the centralized critic is used during training only. The encoder processes the joint observation sequence to form a latent world context. The auto-regressive decoder, utilizing causal masking, generates the action for each agent sequentially, conditioned on the world context and previously generated actions.

$z_{i,t} = \mathcal{M}_\phi(o_{i,t})$. This latent variable encapsulates the learned spatio-temporal dynamics and global statistical correlations acquired during centralized training. The Transformer decoder then generates the action $a_{i,t}$ conditioned on this context:

$$\pi_\theta(a_{i,t}|o_{i,t}) \approx \pi_\theta(a_{i,t}|o_{i,t}, z_{i,t}, a_{<i,t}). \quad (51)$$

To integrate the global context, the actor network employs a cross-attention mechanism where $\mathbf{z}_t$ serves as the Key and Value, and the local observation $o_{i,t}$ serves as the Query. This fusion mechanism enables selective attention to relevant global information based on local task requirements, and the compact 256-dimensional latent vector ensures that the communication overhead remains minimal.

The WM-MAPPO algorithm optimizes the offloading decision variables $\mathbf{X}$ (routing). The computational resource allocation variables $\mathbf{f}$ are determined by a First-Come-First-Served (FCFS) policy. Once a task is routed to a satellite s , it enters the task queue $Q_s(t)$. The satellite allocates its full computational capacity F_s to the task at the head of the queue, i.e., $f_{s,k} = F_s$, until the task is completed. This strategy ensures sequential processing based on arrival order, minimizing the waiting time for the head-of-line task and maintaining fairness, consistent with the queuing delay model.

IV. SIMULATION RESULTS

This section validates the performance of the proposed framework in a multi-layer satellite network. Convergence analysis, performance evaluation, and robustness as well as generalization are demonstrated.

A. Simulation Setup

To rigorously evaluate the performance of the proposed WM-MAPPO framework, a realistic large-scale simulation environment is constructed to emulate an integrated

space-ground network. The environment comprises multiple satellite constellations and terrestrial ground stations interacting dynamically under varying mission workloads [34].

1) *Scenario Description and Network Configuration:* The simulation executes over a maximum simulation time of 200 s, employing a simulation timestep of 0.1 s and orbit propagation interval of 1 s. The simulated network consists of three satellite constellations with distinct functionalities:

- 4,408 LEO communication and computation satellites inspired by the StarLink V1 constellation, orbiting at altitudes of 540 km–570 km with inclinations ranging from 53.0° to 97.6°.
- 100 GEO satellites positioned at approximately 35,786 km altitude for consistent sensing and communication.
- 2,880 SSO remote sensing satellites orbiting at 700 km with a 98.0° inclination.

The satellite constellations are complemented by a terrestrial component of 34 ground stations located strategically in major cities across China. Visibility constraints for satellite-ground connectivity adopt a minimum elevation angle of 25°, and safe inter-satellite link clearance is set to 150 km.

It is noted that while the environment simulates the orbital dynamics of the full constellation to maintain topological fidelity, the number of active MARL agents participating in the collaborative decision-making process is $N = 311$. This subset represents the satellites with visibility to the covered ground stations during the simulation window. This focus ensures computational efficiency while preserving the complexity of large-scale coordination.

2) *Node Resources and Communication Characteristics:* To realistically model computation capacities, LEO satellites feature powerful onboard computational resources comprising

Algorithm 1 WM-MAPPO Offloading Framework

```

1: Initialize: actor-decoder  $\pi_\theta$ , critic  $V_\psi$ , world model
   (encoder)  $\mathcal{M}_\phi$ , replay buffer  $\mathcal{D}$ .
2: Part 1: World Model Encoder Training (Offline)
3: Collect a dataset  $\mathcal{D}_{\text{pre}}$  of trajectories  $(\mathbf{o}_t, \mathbf{a}_t, r_t, \mathbf{o}_{t+1})$ 
   using a random policy.
4: for each pre-training epoch do
5:   Sample a batch of transitions from  $\mathcal{D}_{\text{pre}}$ .
6:   Train the world model encoder  $\mathcal{M}_\phi$  to map the joint
     observation  $\mathbf{o}_t$  to a context-rich latent state represen-
     tation  $\mathbf{z}_t$ .
7:   Update model parameters  $\phi$ .
8: end for
9: Part 2: WM-MAPPO Policy Training (Online)
10: for each training iteration do
11:   Decentralized Rollout & Data Collection
12:   for each time step  $t = 0$  to  $T_{\text{max}} - 1$  do
13:     1. Candidate Selection and Observation Formulation
14:     Each agent  $i$  identifies candidate nodes  $\mathcal{C}_{i,t}$ .
15:     Each agent formulates its local observation  $\mathbf{o}_{i,t}$ .
16:     Form the joint local observation  $\mathbf{o}_t$ .
17:     2. Centralized World Model Prediction
18:     Centrally encode the joint observation into a latent
       state:  $\mathbf{z}_t \leftarrow \mathcal{M}_\phi(\mathbf{o}_t)$ .
19:     Distribute the global latent state  $\mathbf{z}_t$  to all agents.
20:     3. Decentralized Auto-Regressive Policy
21:     Generate joint action  $\mathbf{a}_t$  auto-regressively, where
       each agent's action is conditioned on the predicted
       global latent state and previous agents' actions:
22:      $\mathbf{a}_t \sim \pi_\theta(\cdot|\mathbf{o}_t) = \prod_{i=1}^N \pi_\theta(a_{i,t}|\mathbf{z}_t, \mathbf{a}_{<i,t})$ .
23:     Execute joint action  $\mathbf{a}_t$ , observe global reward  $r_t$ 
       and next joint observation  $\mathbf{o}_{t+1}$ .
24:     Store trajectory fragment  $(\mathbf{o}_t, \mathbf{a}_t, r_t, \mathbf{o}_{t+1})$  in  $\mathcal{D}$ .
25:   end for
26:   Centralized Parameter Update
27:   Sample a batch of trajectories  $\tau$  from  $\mathcal{D}$ .
28:   Compute advantages  $\hat{A}_t^{\text{GAE}}$  and value targets  $G_t$ .
29:   Compute Actor Loss  $\mathcal{L}_{\text{CLIP}}(\theta)$ .
30:   Compute Critic Loss:  $\mathcal{L}_V(\psi) \leftarrow \mathbb{E}[(V_\psi(\mathbf{o}_t) - G_t)^2]$ .
31:   Update actor parameters  $\theta$  by minimizing  $\mathcal{L}_{\text{CLIP}}(\theta)$ .
32:   Update critic parameters  $\psi$  by minimizing  $\mathcal{L}_V(\psi)$ .
33: end for

```

4 CPU cores operating at frequencies of 2.0 to 8.0 GHz (assuming a standard complexity of 17 cycles/byte). The communication sub-system encompasses ISL optical links operating at 193.5 THz (1–4 GHz bandwidth, 0.5–5 W power).

3) *Dynamic Task Generation Mechanism:* To simulate realistic operational demands, a geographically driven stochastic task generation mechanism is adopted. Specifically, task arrivals follow a Poisson distribution and are spatially correlated with real-world data such as population density, GDP, and global tree age distribution. This methodology generates a diverse and challenging workload tailored to represent three distinct application categories, each designed to stress different aspects of the space-based network:

TABLE I
EXPERIMENTAL LOAD LEVEL CONFIGURATION

Load Level	Comm. Demand (MB·s ⁻¹ /person)	Comp. Demand (MB·s ⁻¹ /person)	Sensing Interval (minutes)
1 (Low)	0.0005	0.00006	240
2	0.0010	0.00012	120
3	0.0015	0.00018	90
4 (Mid)	0.0020	0.00024	60
5	0.0025	0.00030	55
6	0.0030	0.00036	45
7 (High)	0.0035	0.00042	35

- **Communication Tasks:** These tasks represent large-scale, intra-constellation data transfers, such as relaying a large scientific data file or a segment of high-definition Earth observation video between satellites. They are defined by a substantial data size ($\sim \mathcal{N}(30000, 10000^2)$ MB) and a stringent time-to-live (TTL) constraint ($\sim \mathcal{U}(10, 20)$ s).
- **Computation Tasks:** These tasks simulate complex, on-orbit data processing workloads, such as running AI inference on an image, data fusion, or signal processing. They are characterized by a moderate data size ($\sim \mathcal{N}(3000, 1000^2)$ MB) and a similar tight TTL ($\sim \mathcal{U}(10, 20)$ s).
- **Remote Sensing Tasks:** These tasks act as the primary source of novel data within the simulation, representing Earth observation requests. They are defined by their required observation duration ($\sim \mathcal{U}(11, 22)$ s) and a specified data quality or precision ($\sim \mathcal{U}(0.2, 0.8)$).

Experiments are conducted across several systematically-derived network load levels. As shown in Table I, these levels are designed by progressively increasing task demand intensities and remote sensing observation frequencies to create distinctly operational conditions.

B. Baseline Algorithms

To provide a comprehensive evaluation, the proposed WM-MAPPO method is benchmarked against heuristic, optimization-based, Lyapunov-based, and learning-based baselines, covering both conventional and DRL-based task offloading strategies.

- *Random Selection:* As a naive benchmark representing a lower bound on performance, this algorithm assigns each incoming task to a randomly selected candidate satellite.
- *Greedy Algorithm:* This algorithm represents a classic, state-aware heuristic. For each task, it calculates an estimated completion time for every possible candidate satellite.
- *Heuristic-Based Algorithm:* This baseline simulates a traditional, rule-based system. It employs a multi-attribute utility function to score each candidate satellite. The score is a weighted sum of normalized metrics, including available computational capacity, current task queue length, and the data rate of the communication link to the node.
- *Lyapunov-DPP Algorithm:* A queue-aware drift-plus-penalty baseline. For each task, it scores feasible

candidates using estimated delay, virtual backlog, deadline risk, and hop count.

- *MINLP-inspired Optimization*: An optimization-based baseline inspired by mixed-integer nonlinear programming. It selects the minimum weighted-cost candidate from simulator-generated feasible routes, considering delay, energy, deadline violation, hop count, resource pressure, and sensing quality.
- *D3QN* [18]: As a representative of advanced single-agent, value-based DRL methods, we adapt the D3QN architecture. This is an individual agent learning paradigm, rendering other agents as part of the environment.
- *COMA* [20]: To benchmark against a classic multi-agent actor-critic algorithm, COMA is implemented and adapted from recent work applying it to space-ground networks. COMA addresses the multi-agent credit assignment problem by computing a counterfactual advantage function for each agent. This baseline represents a foundational MARL approach without explicit world modeling.
- *T-MAPPO*: To isolate the performance gains attributable to the predictive world model from those provided by the Transformer architecture, a T-MAPPO baseline is introduced. T-MAPPO utilizes the same Transformer-based policy architecture as WM-MAPPO but operates without the predictive world model component, serving as the key ablation baseline.

All reinforcement learning-based algorithms were trained using a consistent set of core hyperparameters where applicable, determined through empirical tuning. The PPO-specific parameters for the proposed WM-MAPPO and other relevant policy-gradient baselines include a discount factor (γ) of 0.96, a GAE parameter (λ) of 0.98, and a policy clipping ratio (ϵ) of 0.2. The clipping parameter $\epsilon = 0.2$ imposes a trust region constraint that prevents excessively large policy updates, following the recommendation of the original PPO formulation [35]. The GAE parameter $\lambda = 0.98$ balances the bias-variance tradeoff in advantage estimation, ensuring accurate credit assignment for long-horizon tasks [35]. These values are consistent with established configurations in DRL-based task offloading studies within resource-constrained environments [36]. Common parameters across all DRL baselines include a learning rate of 1×10^{-4} and neural network hidden layers of 512 dimensions. Exploration for all agents is managed via an ϵ -greedy strategy, with the exploration rate ϵ decaying exponentially from an initial value of 0.9 to a final value of 0.01 at a rate of 0.9 per episode. For methods employing a Transformer architecture, including our proposed WM-MAPPO, we configure the model with a latent dimension of 256, 8 multi-head attention heads, and 4 transformer layers. The training is conducted with a batch size of 32 and an observation sequence length of 50.

C. Convergence Analysis

Before evaluating the convergence of the overall MARL agent, we first analyze the training performance of the foundational world model component. Figure 4 illustrates the

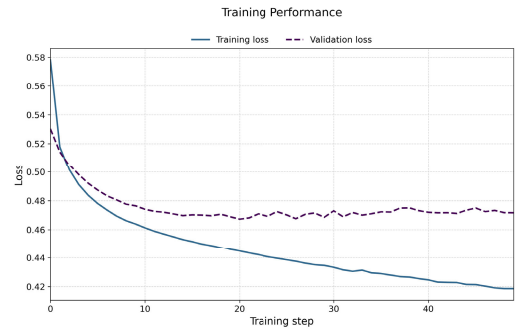


Fig. 4. Training curve of the proposed Transformer-based world model based on sampled trajectories.



Fig. 5. Ablation study on the impact of different learning rates on the convergence performance.

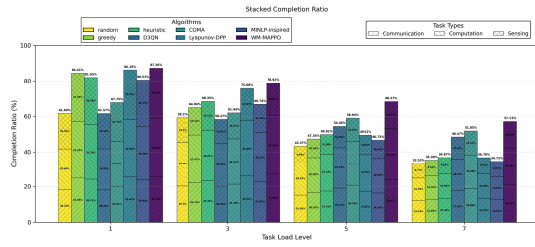
training curve of our Transformer-based world model, plotting its prediction loss on a dataset of trajectories sampled from the environment. The curve demonstrates a rapid and stable convergence, with the loss decreasing sharply during the initial epochs. This indicates that the model quickly learns the fundamental spatio-temporal dynamics of the system.

Then, we evaluate the training stability and hyperparameter sensitivity of our proposed WM-MAPPO algorithm. Figure 5 presents an ablation study on the learning rate, plotting the average standardized episode reward as a function of training episodes to observe its impact on convergence performance. As shown in Fig. 5, the choice of learning rate significantly affects the training process. A relatively high learning rate of 1×10^{-3} proves to be too aggressive, the large policy updates cause the training to become unstable and prematurely converge to a suboptimal policy, indicating that the agent has likely fallen into a local optimum.

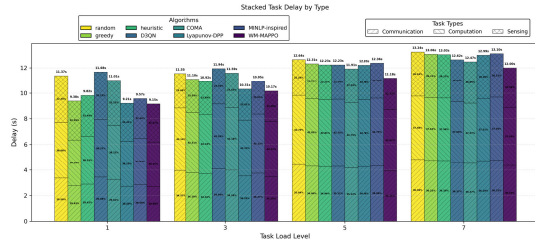
Meanwhile, by creating a rich and predictive latent representation of the environment, the world model enables the agent to extract valuable experience even from complex and noisy state transitions. Consequently, the training performance is not severely hampered by environmental noise, leading to stable learning across a wider range of lower learning rates. Among the successful candidates, the rate of 1×10^{-4} is selected for all subsequent experiments as it provides the optimal balance, achieving the fastest convergence speed without sacrificing the stability and high final reward level demonstrated by the other effective rates.

D. Performance Evaluation

We now conduct a comprehensive performance evaluation of all algorithms across various system load levels, focusing



(a) Overall Completion Ratio by Task Load



(b) Overall Delay by Task Load

Fig. 6. Aggregated performance comparison of all baseline algorithms, showing the composition of (a) task completion ratio and (b) delay across different loads.

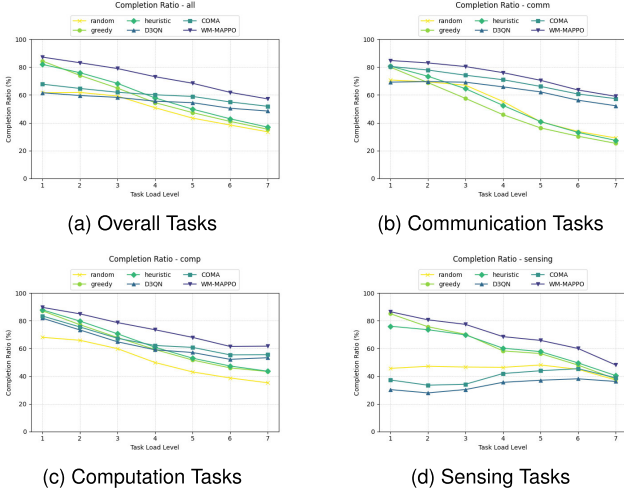


Fig. 7. Task completion ratio of different tasks as a function of increasing system load levels.

on two key metrics: Task Completion Ratio and Delay. This analysis is conducted under a task distribution where the ratio of Communication:Computation:Sensing tasks is approximately 0.37:0.37:0.26. Given that computation tasks have minimal communication requirements, the significant presence of communication-specific tasks creates intense competition for network resources, a challenge that WM-MAPPO manages most effectively.

Fig. 6 shows that WM-MAPPO achieves the highest completion ratio and lowest delay under Loads 1, 3, 5, and 7. The greedy, heuristic, Lyapunov-DPP, and MINLP-inspired baselines remain competitive under light loads, but degrade faster as resource contention increases. COMA and D3QN become more competitive under heavier loads, while WM-MAPPO remains best overall due to predictive latent lookahead.

Figure 7 presents the task completion ratio of each algorithm under different load conditions, providing insights into how effectively each algorithm fulfills incoming tasks as the

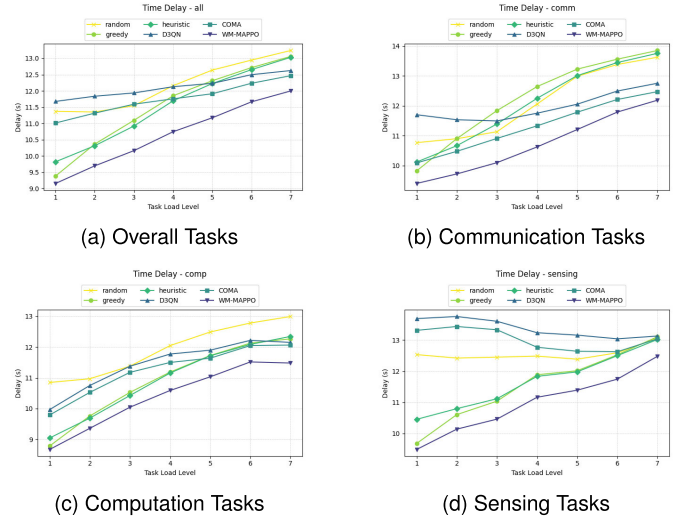


Fig. 8. Average task delay of different tasks as a function of increasing system load levels.

system load increases. As expected, the overall trend clearly demonstrates a decline in task completion ratio with escalating task load, due to the increasing shortage of system resources and heightened task competition. Significantly, the proposed WM-MAPPO algorithm consistently achieves the highest completion ratio across all task load levels and task categories, highlighting its superior ability in resource allocation and task scheduling. In fact, WM-MAPPO's performance remains remarkably stable, exhibiting considerably slower degradation than the other algorithms under heavier loads. This indicates the robustness and adaptability of WM-MAPPO in congested and highly dynamic environments.

To further assess the algorithms' overall efficiency, Fig. 8 illustrates the evolution of average task delay as task load increases. In general, average delays rise with heavier workload demands for communication and computation tasks, reflecting extended queue lengths and intensified resource contention. A notable exception is observed for sensing tasks (Fig. 8 (d)), where the average delay decreases under high load. This is primarily attributed to the strict time-to-live constraints of sensing tasks: under high load, complex sensing tasks with long waiting times fail and are excluded from the average delay calculation, leaving only tasks with immediate visibility windows (a survivor bias effect). Consistent with the task completion ratio results, WM-MAPPO outperforms other algorithms significantly in terms of task processing speed. Specifically, the proposed algorithm maintains the lowest average task delays across all task levels and task types investigated, demonstrating its remarkable capability to rapidly handle incoming tasks, thereby reducing queuing and waiting times. D3QN again exhibits commendable performance, securing the second-lowest delay across most scenarios, while COMA and greedy algorithms exhibit intermediate performance levels. Conversely, heuristic and random strategies produce substantially higher delays, indicating that their relatively simplistic decision-making rules cannot effectively manage task scheduling amidst dynamic resource constraints, thus severely compromising system efficiency.

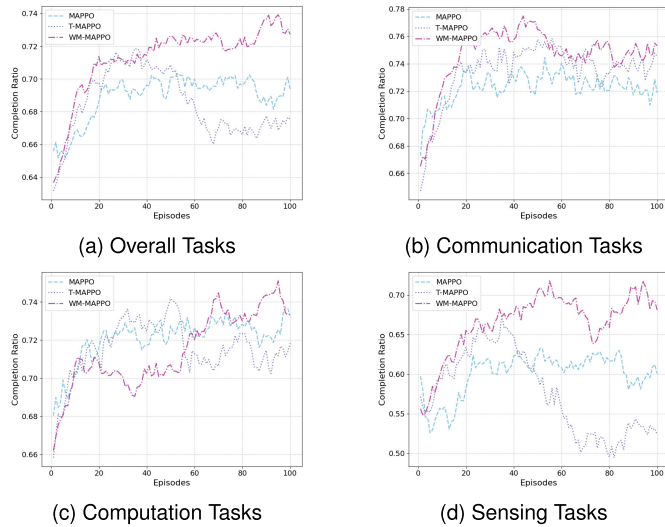


Fig. 9. Ablation study: training curves of WM-MAPPO, T-MAPPO, and MAPPO across different task types.

Among other algorithms, D3QN performs competitively in computation-intensive scenarios but exhibits performance degradation in sensing environments where inter-agent coordination is required. On the other hand, COMA and greedy algorithms deliver moderate performance that deteriorates significantly with increasing load. The heuristic and random strategies demonstrate notably poor performance, with a key exception for sensing tasks where the random algorithm unexpectedly outperforms the heuristic one. This counterintuitive result exposes the limitation of rule-based approaches in multi-task scenarios; the heuristic's fixed rules likely create a detrimental bias against sensing tasks under contention. The random strategy, lacking this systemic bias, performs better by chance, underscoring the insufficient adaptability of these simpler approaches to specific task requirements.

1) *Ablation Study on Predictive World Model*: To validate that the performance gains of the proposed framework stem from the predictive capability of the world model, an ablation study is conducted. As illustrated in Fig. 9, WM-MAPPO consistently outperforms T-MAPPO and MAPPO across all task types. While T-MAPPO shows improvements over standard MAPPO by capturing inter-agent dependencies, it fails to match the performance of WM-MAPPO, which leverages the world model to simulate future network conditions. In sensing tasks (Fig. 9 (d)), T-MAPPO exhibits significant performance degradation and instability as training progresses, whereas WM-MAPPO maintains a stable and high completion ratio. This confirms that the internal world model enables agents to simulate future outcomes and transform decision-making from reactive to proactive.

Fig. 10 provides a direct comparison between WM-MAPPO and the model-free MAPPO baseline. WM-MAPPO achieves a higher completion ratio (0.727 vs. 0.693 overall). The most notable improvement occurs in sensing tasks (0.681 vs. 0.601), where the agent must anticipate future visibility windows. WM-MAPPO also consistently maintains lower delay across all task categories. For communication tasks, the average delay is reduced from 10.99 s to 10.65 s. These results confirm that

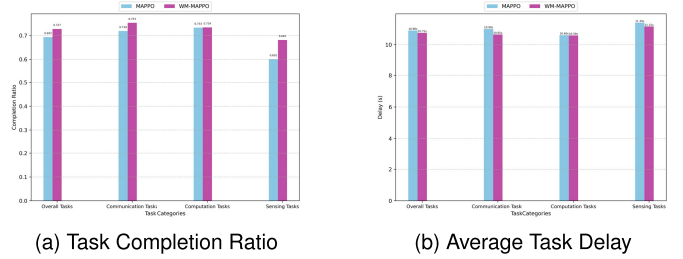


Fig. 10. Performance comparison between WM-MAPPO and the model-free MAPPO baseline across task categories.

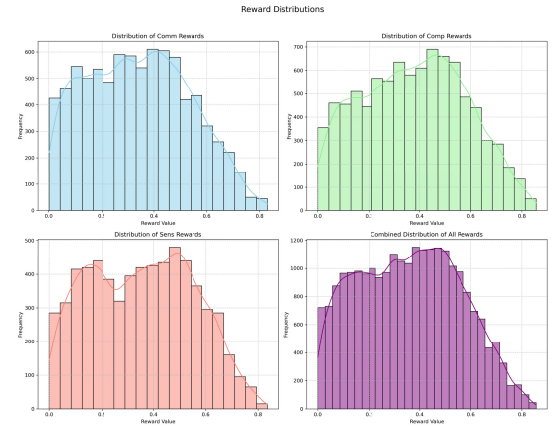


Fig. 11. Empirical distributions of the shared rewards across task categories.

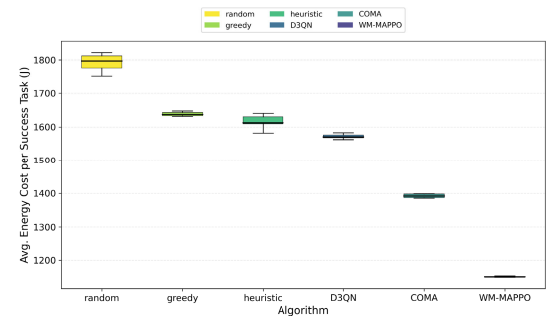


Fig. 12. Boxplot of average effective energy consumption per successful task across different algorithms.

the predictive world model is the key driver of the observed efficiency gains.

2) *Reward Stability Analysis*: To further validate our framework, we investigate the distribution of the shared reward. As illustrated in Fig. 11, the rewards follow a well-behaved distribution concentrated in the $[0, 0.85]$ interval. This dense feedback, combined with our reward normalization and the world model's predictive capability, ensures that the WM-MAPPO algorithm maintains stable convergence and superior performance even under congested network conditions with significant task delays.

3) *Energy Efficiency Analysis*: The energy performance is analyzed using the AEEC metric defined in Section II-E. Fig. 12 presents the boxplot of energy costs per successful task. The proposed WM-MAPPO achieves the lowest median energy cost (approximately 1150 J) and the tightest distribution. Reactive algorithms such as Random and Greedy incur

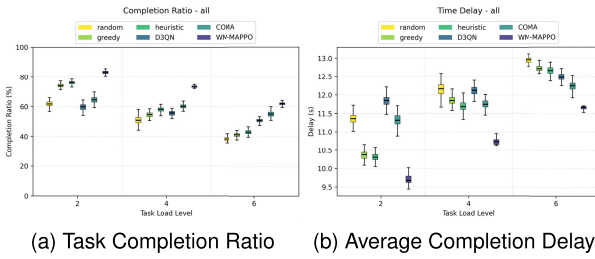


Fig. 13. Performance distribution (mean and variance) of different algorithms on low, medium, and high load scenarios. The DRL-based models were trained only on the medium load.

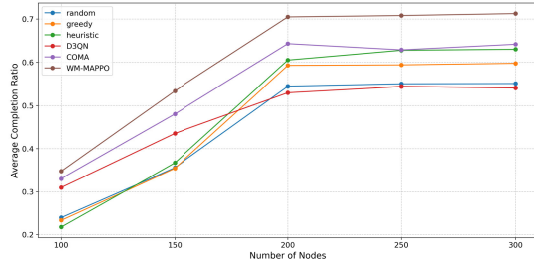


Fig. 14. Performance comparison under random ISL failures across varying network scales.

significantly higher costs (averaging around 1800 J) due to their lack of foresight, which often leads to task failures and wasted energy. Compared to the model-free COMA, WM-MAPPO further reduces energy consumption by leveraging the world model to anticipate network dynamics and select energy-optimal offloading targets.

4) *Computational Cost Analysis*: To assess the real-time feasibility of the proposed WM-MAPPO framework, we analyze computational complexity, training overhead, inference latency, and scalability. For N active agents, latent dimension d , and H_a attention heads, the Transformer encoder has complexity $O(N^2d + Nd^2)$, where self-attention contributes $O(N^2d)$ and the feed-forward layer contributes $O(Nd^2)$. The actor and centralized critic scale as $O(Nd \cdot |A|)$ where $|A|$ denotes the cardinality of the action set and $O(Nd)$ per training step, respectively. In our implementation, the world model converges in 5–10k steps, the full policy stabilizes after about 50k steps with a batch size of 32 and an observation sequence length of 50, and inference for 311 active agents takes about 14 ms within a 100 ms decision interval on an NVIDIA RTX 3090 GPU. Scalability is further supported by Fig. 14, where WM-MAPPO remains stable as active agents increase from 108 to 311 under random ISL failures.

E. Robustness and Generalization

Finally, we assess the generalization capability of our proposed method. The WM-MAPPO agent was trained exclusively under the medium load condition (Load 4). We then evaluated its performance on unseen low and high load scenarios to test its robustness. The performance distribution over 20 evaluation epochs is presented for both completion ratio and completion time in Fig. 13.

The results demonstrate the exceptional robustness and generalization capability of WM-MAPPO. As shown in Fig. 13 (a), despite being trained exclusively on the medium

load scenario, WM-MAPPO not only achieves the highest mean completion ratio on both the unseen low and high loads but also exhibits the smallest variance, indicating high stability.

Complementing this, Fig. 13 (b) illustrates the agent's efficiency, where WM-MAPPO consistently maintains the lowest average task completion time and the tightest performance distribution across all three load levels. This dual success proves that the learned world model successfully captures the underlying system dynamics, allowing the policy to generalize effectively to out-of-distribution conditions and maintain stable, high-quality performance in terms of both success rate and efficiency.

1) *Robustness Under Random ISL Failures*: To further evaluate the resilience of WM-MAPPO in realistic and dynamic environments, random inter-satellite link failures are introduced into the simulation. As shown in Fig. 14, the average completion ratio is evaluated under varying network scales. Despite the increased uncertainty caused by link disruptions, WM-MAPPO demonstrates superior robustness compared to reactive MARL baselines. WM-MAPPO maintains a stable completion ratio of approximately 0.7 even in failure-prone conditions, validating its effectiveness for complex space-centric networking.

2) *Discussion on Environment Dynamics Drift*: Although the world model $\mathcal{M}_\phi$ is trained offline to ensure sample efficiency, its performance under environment dynamics drift is a critical consideration. In practical LEO satellite networks, dynamics drift may arise from hardware aging or unexpected shifts in traffic patterns. The proposed framework mitigates this by learning a robust latent representation of the network state that captures fundamental physical dynamics. Furthermore, the architecture can be extended to incorporate online fine-tuning, where the model parameters are periodically updated using real-time interaction data to track evolving environment dynamics. The experimental results across various load levels (Load 1 to 7) demonstrate that WM-MAPPO possesses strong generalization capabilities, which helps maintain performance even when the deployment environment deviates from the offline training distribution. To further enhance the system's reliability in deployed environments, future work will also focus on uncertainty-aware world models that can detect distribution shifts and trigger online adaptation automatically, ensuring the long-term stability of predictive offloading strategies under significant environment dynamics drift.

V. CONCLUSION

This paper addressed the fundamental challenge of spatio-temporal uncertainty in LEO satellite networks, which limits the effectiveness of traditional, reactive task offloading strategies. The WM-MAPPO framework is introduced as a novel approach that integrates a predictive world model to provide decentralized agents with foresight and lookahead planning capabilities, shifting the decision-making paradigm from reactive to proactive.

The computational complexity of training a global world model could be addressed in future work by investigating more lightweight or decentralized model architectures. Furthermore,

extending the planning horizon by incorporating uncertainty-aware policies could further enhance performance. Expanding the problem formulation to include security constraints would also increase the framework's applicability to real-world operational scenarios.

REFERENCES

- [1] C. Daehnick, I. Klinghoffer, B. Maritz, and B. Wiseman, "Large LEO satellite constellations: Will it be different this time," McKinsey & Company, Chicago, IL, USA, Tech. Rep., 2020.
- [2] M. Hui et al., "A review of LEO-satellite communication payloads for integrated communication, navigation, and remote sensing: Opportunities, challenges, future directions," *IEEE Internet Things J.*, vol. 12, no. 12, pp. 18954–18992, Jun. 2025.
- [3] Fortune Business Insights. (2025). Medium and Large Satellite Market Size. Accessed: May 19, 2025. [Online]. Available: <https://www.fortunebusinessinsights.com/medium-and-large-satellite-market-106278>
- [4] J. Liu, G. Xu, H. Wang, C. Zhang, and X. Lu, "Double λ - κ - μ distribution: Modeling and performance evaluation in non-homogeneous environments," *IEEE Wireless Commun. Lett.*, vol. 14, no. 10, pp. 3378–3382, Oct. 2025.
- [5] F. D. Rossi, P. S. S. D. Souza, and M. C. Luizelli, "Resource allocation on low-Earth orbit edge infrastructure: Taxonomy, survey, and research challenges," *IEEE Access*, vol. 13, pp. 40992–41010, 2025.
- [6] D. Xu et al., "A survey of opportunistic offloading," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 3, pp. 2198–2236, 3rd Quart., 2018.
- [7] W. Stock, R. T. Schwarz, C. A. Hofmann, and A. Knopp, "Survey on opportunistic PNT with signals from LEO communication satellites," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 1, pp. 77–107, Feb. 2025.
- [8] E. Kaya, B. Gorkemli, B. Akay, and D. Karaboga, "A review on the studies employing artificial bee colony algorithm to solve combinatorial optimization problems," *Eng. Appl. Artif. Intell.*, vol. 115, Oct. 2022, Art. no. 105311.
- [9] A. S. Alfa, B. T. Maharaj, S. Lall, and S. Pal, "Mixed-integer programming based techniques for resource allocation in underlay cognitive radio networks: A survey," *J. Commun. Netw.*, vol. 18, no. 5, pp. 744–761, Oct. 2016.
- [10] Q. Chen, Z. Guo, W. Meng, S. Han, C. Li, and T. Q. S. Quek, "A survey on resource management in joint communication and computing-embedded SAGIN," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 3, pp. 1911–1954, Jun. 2025.
- [11] S. Mahboob and L. Liu, "Revolutionizing future connectivity: A contemporary survey on AI-empowered satellite-based non-terrestrial networks in 6G," *IEEE Commun. Surveys Tuts.*, vol. 26, no. 2, pp. 1279–1321, 2nd Quart., 2024.
- [12] F. Tang, H. Hofner, N. Kato, K. Kaneko, Y. Yamashita, and M. Hangai, "A deep reinforcement learning-based dynamic traffic offloading in space-air-ground integrated networks (SAGIN)," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 1, pp. 276–289, Jan. 2022.
- [13] W. Wen, H. Cui, and T. He, "Multi-layer reinforcement learning assisted task offloading in satellite edge computing," *IEEE Trans. Veh. Technol.*, vol. 74, no. 4, pp. 6561–6572, Apr. 2025.
- [14] J. Shuai, H. Cui, Y. He, and M. Guizani, "Dynamic satellite edge computing offloading algorithm based on distributed deep learning," *IEEE Internet Things J.*, vol. 11, no. 16, pp. 27790–27802, Aug. 2024.
- [15] B. Xie, H. Cui, P. Cao, Y. He, and M. Guizani, "Computation offloading optimization in satellite-terrestrial integrated networks via offline deep reinforcement learning," *IEEE Internet Things J.*, vol. 11, no. 23, pp. 38803–38814, Dec. 2024.
- [16] Y. Hou, Z. Wei, R. Zhang, X. Cheng, and L. Yang, "Hierarchical task offloading for vehicular fog computing based on multi-agent deep reinforcement learning," *IEEE Trans. Wireless Commun.*, vol. 23, no. 4, pp. 3074–3085, Apr. 2024.
- [17] Z. Wei, B. Li, R. Zhang, X. Cheng, and L. Yang, "OCVC: An overlapping-enabled cooperative vehicular fog computing protocol," *IEEE Trans. Mobile Comput.*, vol. 22, no. 12, pp. 7406–7419, Dec. 2022.
- [18] J. Wu, M. Jia, N. Zhang, and Q. Guo, "Multi-agent deep reinforcement learning-based computation offloading in LEO satellite edge computing system," *IEEE Commun. Lett.*, vol. 28, no. 10, pp. 2352–2356, Oct. 2024.
- [19] J. Zhou, J. Liang, L. Zhao, S. Wan, H. Cai, and F. Xiao, "Latency-energy efficient task offloading in the satellite network-assisted edge computing via deep reinforcement learning," *IEEE Trans. Mobile Comput.*, vol. 24, no. 4, pp. 2644–2659, Apr. 2025.
- [20] H. Zhang, H. Zhao, R. Liu, A. Kaushik, X. Gao, and S. Xu, "Collaborative task offloading optimization for satellite mobile edge computing using multi-agent deep reinforcement learning," *IEEE Trans. Veh. Technol.*, vol. 73, no. 10, pp. 15483–15498, Oct. 2024.
- [21] J. Lai, H. Liu, Y. Sun, H. Tan, L. Gan, and Z. Chen, "Multi-agent deep reinforcement learning aided computing offloading in LEO satellite networks," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2023, pp. 3438–3443.
- [22] S. Fan and J. Zhang, "Multi-agent deep reinforcement learning for task offloading in LEO satellite networks," in *Proc. 10th Int. Conf. Comput. Commun. Syst. (ICCCS)*, Apr. 2025, pp. 698–703.
- [23] B. Li, K. Chen, Z. Song, R. Zhang, and X. Cheng, "Divide and conquer: Advancing large-scale multi-agent pathfinding with hierarchical reinforcement learning," *IEEE Trans. Mobile Comput.*, vol. 25, no. 1, pp. 102–116, Jan. 2026.
- [24] J. Lai, H. Liu, Y. Sun, J. Zhu, W. Ma, and L. Gan, "Multi-agent deep reinforcement learning based computation offloading approach for LEO satellite broadband networks," in *Proc. IEEE Symp. Comput. Commun. (ISCC)*, Jul. 2023, pp. 1435–1440.
- [25] S. S. Hassan, Y. M. Park, Y. K. Tun, W. Saad, Z. Han, and C. S. Hong, "Satellite-based ITS data offloading & computation in 6G networks: A cooperative multi-agent proximal policy optimization DRL with attention approach," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 4956–4974, May 2024.
- [26] J.-H. Chen, W.-C. Kuo, and W. Liao, "SpaceEdge: Optimizing service latency and sustainability for space-centric task offloading in LEO satellite networks," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 15435–15446, Oct. 2024.
- [27] M. Jia, L. Zhang, J. Wu, Q. Guo, G. Zhang, and X. Gu, "Deep multiagent reinforcement learning for task offloading and resource allocation in satellite edge computing," *IEEE Internet Things J.*, vol. 12, no. 4, pp. 3832–3845, Feb. 2025.
- [28] S. Zhang, A. Liu, C. Han, X. Liang, X. Xu, and G. Wang, "Multiagent reinforcement learning-based orbital edge offloading in SAGIN supporting Internet of Remote Things," *IEEE Internet Things J.*, vol. 10, no. 23, pp. 20472–20483, Dec. 2023.
- [29] C. Yu et al., "The surprising effectiveness of PPO in cooperative, multi-agent games," 2021, *arXiv:2103.01955*.
- [30] Y. Zhang and D. Guo, "Multi-agent reinforcement learning for multi-cell spectrum and power allocation," 2023, *arXiv:2312.05746*.
- [31] T. M. Moerland, J. Broekens, A. Plaat, and C. M. Jonker, "Model-based reinforcement learning: A survey," *Found. Trends Mach. Learn.*, vol. 16, no. 1, pp. 1–118, Jan. 2023.
- [32] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," 2017, *arXiv:1701.01090*.
- [33] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*.
- [34] Z. Wei, B. Li, R. Zhang, X. Cheng, and L. Yang, "AirFogSim: A light-weight and modular simulator for UAV-integrated vehicular fog computing," *IEEE Trans. Mobile Comput.*, vol. 25, no. 5, pp. 6755–6766, Dec. 2025.
- [35] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," 2015, *arXiv:1506.02438*.
- [36] Z. Wei, B. Li, R. Zhang, X. Cheng, and L. Yang, "Many-to-many task offloading in vehicular fog computing: A multi-agent deep reinforcement learning approach," *IEEE Trans. Mobile Comput.*, vol. 23, no. 3, pp. 2107–2122, Mar. 2024.



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